Project 41 // Review 2

Campus Facility Maintenance

3NF design, constraints & triggers, SQL reports, prototype

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Problem & Structural Scope

Database-driven Workflow

Replacing fragile manual registers.

16 Tables // 4 Domains

Campus

Building, Room, Asset, Campus_User

Reference

Category, Priority, Technician, Status_Transition

Workflow

Maintenance_Request, Assignment, Work_Log, Status_History, Feedback

Cost & Stock

Material, Material_Usage, Request_Cost

3NF Normalization Strategy

Functional Dependencies

RequestID → RequestNo, RoomID, CategoryID, PriorityID, Status ...

RequestNo → RequestID (Alt Key)

RoomID → BuildingID, Room_No

Material_ID → Material_Name

- ✓ Building is derived, not stored.
- ✓ Material cost is derived.
- ✓ Material_Usage keeps price snapshot (historical fact).

Data Dictionary: Request

Column	Type	Key	Null	Constraint
RequestID	INT	PK	No	AUTO_INCREMENT
RequestNo	VARCHAR	UK	No	UNIQUE, CHECK format
RoomID	INT	FK	No	Ref: Room
Status	VARCHAR	-	No	DEFAULT 'Open'
Remarks	TEXT	-	Yes	Trigger: Req if Closed

Comprehensive dictionaries established for all 16 tables.

DDL Integrity Rules

- Unique request numbers (UNIQUE + format CHECK).
- Non-negative costs, quantities, stock (CHECK).
- Active technician under MaxActiveJobs limits.
- Closure details required (CHECK + trigger).
- Valid status transitions via mapping table trigger.
- **9 Triggers deployed.**

Status Flow Machine



↺ Reopen loop: Resolved → Open

Side states: On Hold, Cancelled

Trigger validation: 12 legal transitions out of 49.

Analytical SQL Reports

1. **Pending requests:** View, JOINs, CASE.
2. **Ageing buckets:** Derived table classification.
3. **Response vs Target:** Subquery conditional aggregate.
4. **Technician load:** View, aggregated LEFT JOINs.
5. **Building-wise cost:** Correlated subquery, HAVING.
6. **Material use:** RANK() window function.

Feedback View using correlated subqueries.

Sample Results Snapshot

12 OCT 2026 | N=21 Requests

9
Pending Requests

1
SLA Breach (16h vs 4h)

₹5.4k
Costliest (Admin/Aud)

Response vs Target

Crit/Low: 100%

High: 83%

Medium: 86%

Operational Highlights

⚠️ Projector Lamp flagged REORDER.

⭐ Ratings span 3.5 (Furniture) to 5.0.

🔥 Critical breach: Kitchen exhaust fan.

Prototype & Roadmap

Simulator: Triggers Blocked

- Close without remarks.
- Skip Open → Resolved.
- Assign at max limit.

Review 3 Target

- Python application integration.
- Stored procedures & transactions.
- Validation & error handling.
- Indexing & testing.

Thank you. Questions?